Charging Basics

Home Charging

Plug your vehicle into a Level 2 home charger when you return from a trip or during the day. A standard 240-volt circuit with a 40-amp breaker and a 240V/40A NEMA 14-50 outlet is recommended for optimal charging. With this setup, you can add up to 50 miles of range per hour. For example, a 40-mile drive will take about 48 minutes to fully recharge from 20% to 100%. Always use the Tesla Wall Connector or a certified third-party Level 2 charger compatible with the vehicle's charging port. Ensure the charging cable is securely connected to both the vehicle and the wall outlet. The charge port light will turn green when charging begins and will remain green until the charge is complete.

Supercharger Use

Superchargers are designed for fast, long-distance travel. Locate one using the vehicle's touchscreen or the Tesla app. When you arrive, park completely within the designated charging spot. Connect the Supercharger cable to the charge port—no need to open the port manually. The vehicle will automatically begin charging at up to 250 kW, depending on battery state and temperature. For instance, a 150-mile range boost can be achieved in about 15 minutes. Charging speed slows as the battery reaches 80% to preserve longevity. You can stop charging at any time by pressing the charge button on the touchscreen or pulling the cable from the port. Always leave enough space around your vehicle to allow others to access their chargers.

Charge Port Will Not Open

If the charge port does not open when you press the charge port button on the touchscreen, first check that the vehicle is unlocked and the key fob is within range. Ensure the charge cable is not connected and that the port is not blocked by debris. Try using the physical charge port release lever inside the driver's side door panel. If the port still does not open, restart the vehicle by turning it off and back on. If the issue persists, contact Tesla Support through the app or website. Do not attempt to force the port open with tools, as this may damage the mechanism.

Charging Stops Early

Charging may stop before reaching your target charge level due to several factors. The vehicle may automatically pause charging if the battery temperature is too high or too low—this typically happens during extreme weather. For example, in temperatures below 32°F (0°C), charging may slow or pause until the battery warms up. Charging may also stop if the power supply is insufficient or if the charging station detects a fault. Check the charging status on the touchscreen. If the charge stops unexpectedly, disconnect and reconnect the cable. If the problem continues, contact Tesla Support for diagnostics.

Setting a Charge Limit

To preserve battery health and avoid overcharging, set a charge limit using the touchscreen. Go to Controls > Battery > Charge Limit. You can choose from 50%, 75%, 80%, 90%, or 100%. For daily use, we recommend setting the limit to 80%—this maintains optimal battery longevity and ensures the vehicle is ready for most trips. If you're planning a long journey, set the limit to 100% before departure.

The vehicle will not charge beyond the set limit, even if the battery is below that level. The charge limit is saved automatically and applies every time you plug in.

Charging Speed

Charging speed depends on several factors. At home, a Level 2 charger adds approximately 30 to 50 miles of range per hour, depending on the charger’s amperage and your vehicle’s state of charge. Supercharging delivers up to 250 kW, enabling a 150-mile range boost in 15 minutes when the battery is below 20%. Speed decreases after 80% to protect the battery. For example, going from 20% to 80% takes about 25 minutes at a Supercharger. Charging speed also drops in cold weather—below 32°F (0°C), charging may be slower and limited to 100 kW or less until the battery warms. Always use the correct charging equipment and ensure the vehicle is in a well-ventilated area.